- **2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem**
  - **Analysis:** This is the most critical ability for this question. Students need to look at the trends in the bar chart and evaluate the reasonableness of the statement, "By about 2015 these three penguin types will be extinct." The chart shows that the average birth rate fluctuates between approximately 0.4 and 1.2 chicks per couple, with no clear long-term downward trend towards zero. Therefore, judging the conclusion of "extinction" as unreasonable based on the chart data is necessary.
- **1. Interpret a mathematical result back into the real world context**
  - **Analysis:** Students must be able to translate the numerical values in the chart (e.g., an average of around 1.1 chicks in 2000) into an understanding of the real-world situation (the survival status of the penguin population). This involves extracting meaning from the mathematical representation (the chart) and applying it to the real-world context.
- **5. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation**
  - **Analysis:** Students need to read and interpret the bar graph as a form of mathematical representation, including data for three different species annually, to compare trends and assess the truthfulness of the statements.
- **7. Construct and communicate explanations and arguments in the context of the problem**
  - **Analysis:** Solving the problem requires providing reasons for the judgment (true/false), for example, explaining why the data shown in the chart indicates a relatively stable population rather than one heading toward extinction.

- **2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem** (p. 1)
  - **分析：** 这是该问题最核心的能力。学生需要查看图表中的趋势，并评估“到 2015 年这三种企鹅将灭绝”这一陈述的合理性。图表显示企鹅的平均生育率在 0.4 到 1.2 只幼崽之间波动，并且没有明显的长期下降趋势趋于零。因此，根据图表数据判断“灭绝”的结论**不合理**。
- **1. Interpret a mathematical result back into the real world context** (p. 1)
  - **分析：** 学生必须能够将图表中的数值（例如，2000 年的平均幼崽数量约为 1.1 只）转化为对现实世界情境（企鹅种群生存状态）的理解。这涉及从数学表示（图表）中提取含义并将其应用于现实背景。
- **5. Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation** (p. 1)
  - **分析：** 学生需要阅读并解释条形图这种数学表示形式，包括每年三个不同物种的数据，以便比较趋势并评估陈述的真伪。
- **7. Construct and communicate explanations and arguments in the context of the problem**
  - **分析：** 解决问题需要对判断（真/假）给出理由，例如解释为什么图表中的数据显示种群数量相对稳定，而不是走向灭绝。